SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 8

Risk & Return and the Cost of Capital (WACC)

This week answers two big corporate-finance questions. First: what return is fair for a given risk? We compute return, risk (standard deviation and coefficient of variation), the diversification effect in a portfolio, beta, and the CAPM with the Security Market Line. Second: what does the firm's money cost? We compute the cost of debt (before and after tax), preferred stock, and common equity (Gordon model and CAPM), then combine them into the WACC — the minimum return (hurdle rate) a project must beat.

Material

1 Roadmap: two parts, one thread

Part 1 · Risk & Return (Ch. 8)
Single-asset return and risk → portfolio risk (correlation and diversification) → beta, CAPM, Security Market Line.

Part 2 · Cost of Capital (Ch. 9)
Cost of long-term debt, preferred stock, common equity → WACC as the hurdle rate.

Investors bear risk → They require a return (CAPM)

→ Required return = the firm's cost → Combined → WACC

→ Projects must beat WACC

The thread: **the return investors require is the firm's cost of capital**. Riskier firm → higher required return → more expensive capital → harder for projects to pass.

2 Fundamentals: portfolio, risk and total return

Term	Meaning
Portfolio	A collection or group of assets.
Risk	A measure of the *uncertainty* surrounding the return an investment will earn.
Total rate of return	Total gain or loss over a period as a percentage of the starting value: cash distributions plus change in value, divided by beginning value.

$$r_t = (C_t + P_t - P_{t-1}) \div P_{t-1} \quad (8.1)$$

≡ Example 8.1 — Robin, 2016

Stock	Start	Dividend	End	Return
Apple	$105.35	$2.37	$115.82	(2.37 + 115.82 − 105.35) ÷ 105.35 = **12.2%**
Wal-Mart	$61.46	$2.00	$69.12	(2.00 + 69.12 − 61.46) ÷ 61.46 = **15.7%**

Return includes *both* cash (dividends) and price change (capital gain).

3 Single-asset risk: expected return, standard deviation, coefficient of variation

Standard deviation (σ) is the most common risk indicator: how widely returns *spread* around the average.

$$\text{Expected return: } \bar{r} = \Sigma\, r_j \times Pr_j \quad (8.2)$$

$$\sigma = \sqrt{[\ \Sigma\, (r_j - \bar{r})^2 \times Pr_j\ ]} \quad (8.3)$$

$$\text{From historical data: } \sigma = \sqrt{[\ \Sigma\, (r_j - \bar{r})^2 \div (n - 1)\ ]} \quad (8.3a)$$

≡ Example: assets A and B (Tables 8.3 & 8.4)

Scenario	Prob.	A	B
Pessimistic	0.25	13%	7%
Most likely	0.50	15%	15%
Optimistic	0.25	17%	23%
Expected		**15%**	**15%**
σ		**1.41%**	**5.66%**

Same expected return, but B is far more dispersed → riskier.

Coefficient of variation (CV) measures *relative* risk — risk per unit of return; use it when expected returns differ.

$$CV = \sigma \div \bar{r} \quad (8.4)$$

Example 8.7: $CV_A = 1.41\% \div 15\% =$ **0.094**; $CV_B = 5.66\% \div 15\% =$ **0.377**. B is riskier per unit of return.

4 Portfolio risk: correlation and diversification

An **efficient portfolio** maximises return for a given level of risk.

$$r_p = \Sigma\, w_j \times r_j \quad (8.5)$$

Portfolio return is always a weighted average — portfolio risk is **not**, because it depends on correlation.

Term	Meaning
Correlation	Statistical measure of the relationship between two series.
Positive	Series move in the same direction.
Negative	Series move in opposite directions.

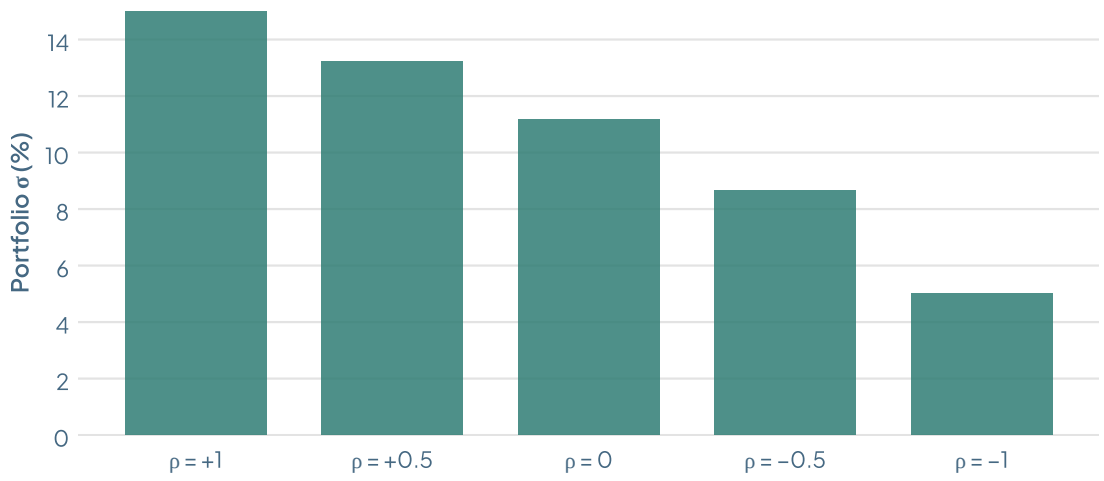
Correlation coefficient (ρ)	From +1 (perfect positive) to −1 (perfect negative).

Diversification rules

- Combine assets with the **lowest possible correlation**.
- Low correlation reduces the variability of portfolio returns.
- With **perfectly negative** correlation, some mix makes the portfolio **risk-free**.

Supplement (standard formula, not on the slides): $\sigma_p = \sqrt{(w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2)}$. Example: 50/50 in assets with σ = 10% and 20%.

Risk of a 50/50 portfolio (σ1 = 10%, σ2 = 20%) by correlation



Return stays the weighted average, but risk falls from 15% to 5% as correlation turns negative; with 2/3–1/3 weights, ρ = −1 gives σ = 0.

5 Two kinds of risk: diversifiable and non-diversifiable

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Total risk = Nondiversifiable risk + Diversifiable risk   (8.6)
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Diversifiable (unsystematic)

Firm-specific random causes: strikes, lawsuits, a CEO leaving, a

vs

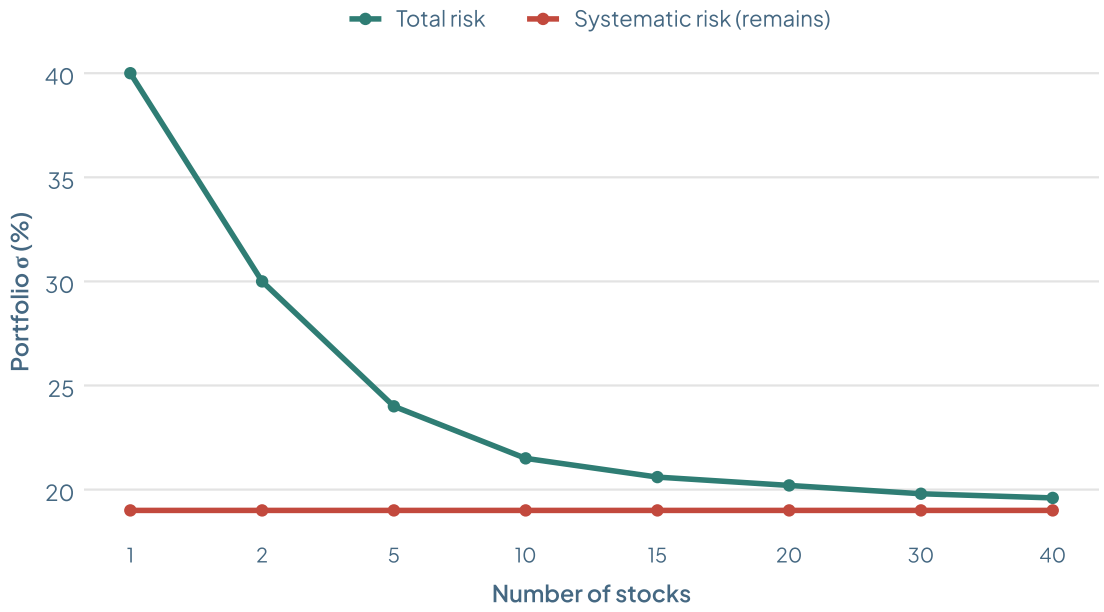
Nondiversifiable (systematic)

Market factors affecting all firms: interest rates, inflation, recession,

failed product. **Can be eliminated** by diversification.

war, policy. **Cannot be eliminated** — the *relevant* risk.

Diversification: portfolio risk vs number of stocks (illustrative)



The gap between the lines is unsystematic risk removed by diversification; beyond ~15–20 stocks the gain is small.

As more (imperfectly correlated) stocks are added, unsystematic risk disappears, leaving only market risk. Since unsystematic risk can be removed for free, the market **doesn't reward** bearing it — only systematic risk is rewarded. That is the basis of the CAPM.

6 Beta and the CAPM

The **CAPM** is the classic theory linking nondiversifiable risk to expected (required) return for all assets.

Beta (β) = a relative measure of systematic risk: how much an asset's return moves when the market return changes. **Market return** = return on the portfolio of all traded securities.

β	Meaning	Reading

$\beta = 1.0$	Moves with the market (market & average beta = 1.0)	Market +10% → stock ≈ +10%
$\beta > 1$	More aggressive	$\beta = 2$: market +10% → ≈ +20%
$0 < \beta < 1$	More defensive	$\beta = 0.5$: market −10% → ≈ −5%
$\beta < 0$	Moves opposite (rare)	Market up → asset tends down

Betas can be positive or negative, but positive is the norm; most fall between **0.5 and 2.0**.

$$r_j = R_F + \beta_j \times (r_m - R_F) \quad (8.8)$$

R_F = risk-free rate (usually the U.S. T-bill return); $(r_m - R_F)$ = **market risk premium**.

≡ **Example 8.14 — Benjamin Corporation**

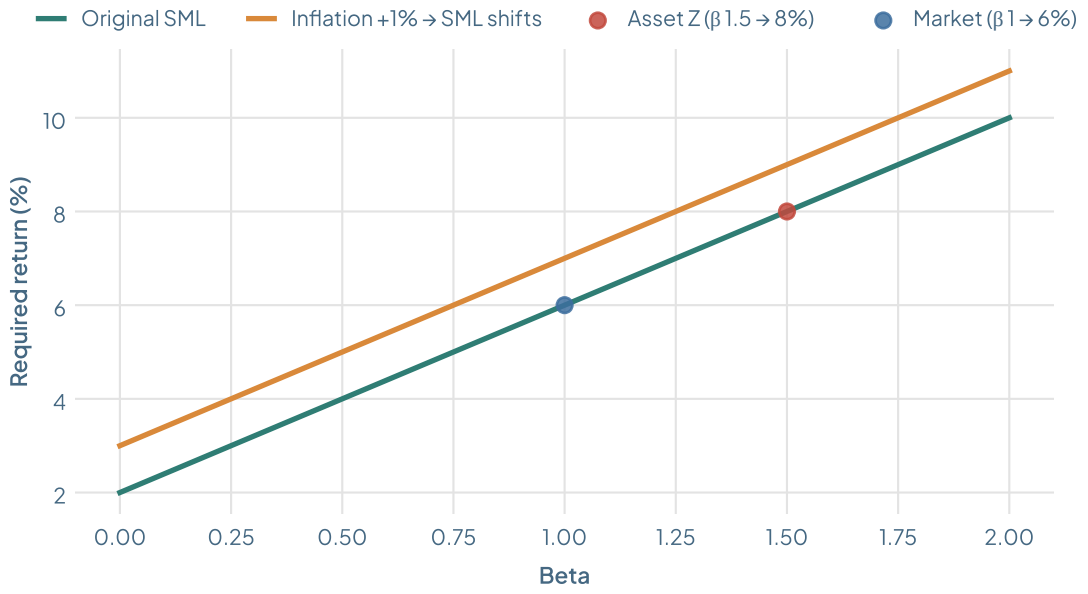
Asset Z: $\beta = 1.5$; $R_F = 2\%$; $r_m = 6\%$.

$r_Z = 2\% + [1.5 \times (6\% - 2\%)] = 2\% + 6\% = \mathbf{8\%}$.

7 The Security Market Line (SML) and inflation

Plotting the CAPM — x = beta, y = required return — gives a straight line called the **Security Market Line (SML)**. Intercept = R_F; slope = market risk premium.

Security Market Line: risk-free 2%, market 6% (Example 8.14)



Intercept = R_F, slope = market premium (4%); higher inflation raises R_F and shifts the line in parallel.

Reading the SML

Assets **above** the line offer more than required (undervalued); **below** = overpriced.

Inflation shifts the SML (Fig. 8.10)

Higher expected inflation raises R_F, shifting the **whole SML up in parallel**.

Extra

If risk aversion rises, the **slope** steepens (bigger market premium), not the intercept.

8 Cost of capital: the basic concept

Term	Meaning
Cost of capital	The firm's cost of financing; the **minimum return** a project must earn to increase firm value.
Capital	Long-term financing: debt and equity.
Capital structure	The firm's mix of debt and equity.

WACC	Weighted average of the costs of debt and equity, weighted by each source's share.

Example 9.1 — judging by one source

	Investment A	Investment B
Expected return	7%	12%
Analyst's call	Accept — "we borrow at 6%"	Reject — "shareholders want 14%"
vs WACC 10%	**Reject**	**Accept**

WACC with a 50–50 mix: (0.50 × 6%) + (0.50 × 14%) = **10%**.

9 Cost of long-term debt

Term	Meaning
Cost of long-term debt	Financing cost of new long-term borrowing.
Net proceeds	Funds actually received from selling a security.
Flotation costs	Total costs of issuing and selling a security.

Example 9.2 — Duchess Corporation

$10m of 20-year, 6% coupon bonds, $1,000 par, sold at par (YTM 6%). Flotation 2% = $20 → **net proceeds $980**.

$$r_d \approx [\ I + (\$1{,}000 - N_d) \div n\ ] \div [\ (N_d + \$1{,}000) \div 2\ ] \quad (9.1)$$

Example 9.4

r_d = ($60 + $1) ÷ $990 = **6.162%**; the exact calculator value is **6.177%**.

$$\text{After-tax cost of debt} = r_d \times (1 - T) \quad (9.2)$$

Example 9.5

21% tax (U.S. after the 2017 TCJA; previously top rate 35%): 6.177% × 0.79 = **4.88%**. If interest already reaches 30% of EBIT, extra interest isn't deductible → before- and after-tax costs are equal.

Why debt is usually cheapest

1. It's **less risky** for investors → lower required return.
2. It has a **tax benefit** equity lacks.

10 Cost of preferred stock

Preferred shares usually pay a **fixed dividend** with a fixed par. Preferred dividends are *not* tax-deductible, so no (1 – T) adjustment.

$$r_p = D_p \div N_p \quad (9.3)$$

Example 9.7 — Duchess

1. Dividend: 8% × \$80 = **\$6.40**
2. Net proceeds: \$80 – \$2.50 = **\$77.50**
3. r_p = \$6.40 ÷ \$77.50 = **8.258%**

11 Cost of common equity: Gordon, CAPM, new issues, retained earnings

The **cost of common equity** = required return on the stock *without* flotation costs; so it equals the cost of retained earnings, while new common stock costs more.

Method 1 · Gordon (constant growth)

Method 2 · CAPM

$$r_s = D_1 \div P_0 + g \quad (9.5)$$

$$r_s = R_F + \beta \times (r_m - R_F) \quad (9.6)$$

Example 9.8 — Gordon

Last dividend \$3.80, growth ≈ 5% → D_1 = \$3.99; P_0 = \$50 → r_s = 0.0798 + 0.05 = **12.98%**.

Example 9.9 — CAPM

R_F 3%, β 1.5, r_m 9% → r_s = 3% + 9% = **12.0%**.

New issue: flotation costs and underpricing lower proceeds and raise the cost.

$$r_n = D_1 \div N_n + g \quad (9.7)$$

Illustration (not from the slides): sell at \$47 with \$2.50 flotation → N_n = \$44.50; r_n = 3.99 ÷ 44.50 + 5% = **13.97%**.

$$\text{Retained earnings: } r_r = r_s \quad (9.8)$$

Typical cost ranking

After-tax debt < preferred < retained earnings (= r_s) < new common (r_n).

12 Weighted Average Cost of Capital (WACC)

$$r_{WACC} = (w_d \times r_d)(1 - T) + (w_p \times r_p) + (w_s \times r_{s \text{ or } n}) \quad (9.9)$$

Key points

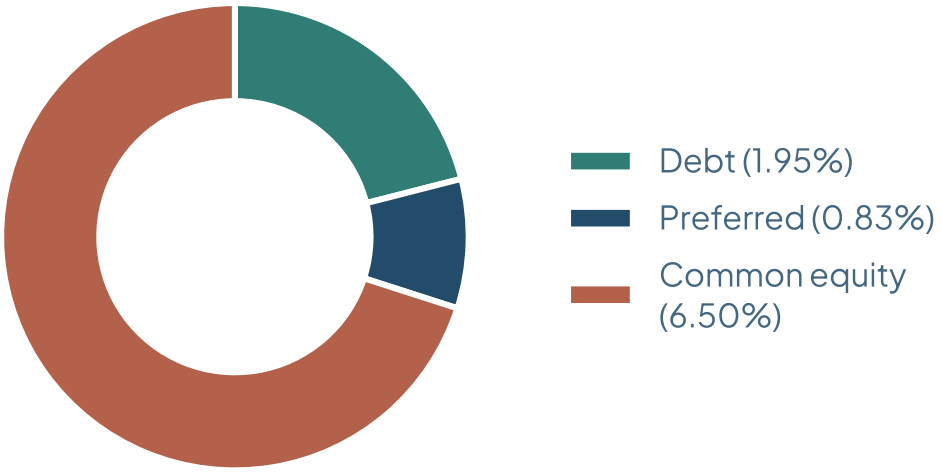
- Weights are nonnegative and sum to 1.0.
- Use **market values** of each source.
- Equity weight × r_s (retained earnings) *or* r_n (new stock).
- Debt cost × (1 – T) for the interest tax deduction.

Example 9.12 / Table 9.2 — Duchess WACC

Source	w	r	w × r
Long-term debt	0.40	4.88%	1.95%
Preferred stock	0.10	8.26%	0.83%
Common equity	0.50	13.00%	6.50%
Total	**1.00**		**WACC = 9.28%**

Average-risk projects must earn **more than 9.28%** to add value.

Each source's contribution to Duchess's 9.28% WACC



Debt is 40% of funding but only ~21% of WACC because it is cheapest.

Weights	Note
Market value weights	Use market values, not book or par.
Target capital structure	The long-run desired debt–equity mix, ideally the firm's optimal mix.

Key takeaways

1. Total return = (cash received + ending price − beginning price) ÷ beginning price.
2. Risk = uncertainty of return, measured by σ; CV = σ ÷ r̄ compares relative risk.
3. Portfolio return is a weighted average; portfolio risk depends on correlation. Lower correlation → better diversification; ρ = −1 can make it risk-free.
4. Total risk = systematic (market, nondiversifiable, relevant) + unsystematic (firm-specific, diversifiable).
5. Beta measures relative systematic risk; market beta = 1.0; most betas are 0.5–2.0.
6. CAPM: $r = R_F + \beta(r_m - R_F)$; plotted as the SML; higher inflation shifts the SML up in parallel.
7. Cost of capital = the minimum project return that increases firm value.
8. After-tax cost of debt = $r_d \times (1 - T)$; debt is usually cheapest (less risky, tax benefit).
9. $r_p = D_p \div N_p$; $r_s = D_1 \div P_0 + g$ (Gordon) or CAPM; $r_n = D_1 \div N_n + g$; $r_r = r_s$.
10. WACC = Σ weight × cost using market-value weights summing to 1; accept projects whose return exceeds WACC.

Case studies

Case & worked answers

Integrated practice: the WACC of PT Kopi Nusantara

Illustrative practice, not a real case or from the slides. PT Kopi Nusantara evaluates 20 new outlets with an 11% expected return. Bonds: par IDR 1,000,000, 8% coupon, 10 years, net proceeds 960,000; tax 22%. Preferred: dividend 9,000, net proceeds 100,000. Equity: R_F 6%, β 1.2, r_m 12%, funded by retained earnings. Target weights: debt 30%, preferred 10%, equity 60%.

1. $r_d = 84{,}000 \div 980{,}000 =$ **8.57%**

2 After tax = 8.57% × 0.78 = **6.69%**

3 r_p = **9.00%**

4 r_s = 6% + 1.2 × 6% = **13.20%**

5 WACC = 2.01% + 0.90% + 7.92% = **10.83%**

Decision: 11% > 10.83% → accept, with a thin margin. Issuing new stock (higher r_n) could push WACC above 11% and flip the decision.

Practice questions

Try answering first. The answer key is below each question.

1. A stock bought at 1,000 pays a 50 dividend and ends at 1,100. Its total return is…

A. 5%
B. 10%
C. 15%
D. 11%

Answer: C. (50 + 1,100 – 1,000) ÷ 1,000 = 15%: dividend plus capital gain.

2. Assets A and B both expect 15%; σ A = 1.41%, σ B = 5.66%. Conclusion?

A. A is riskier
B. B is riskier
C. Same risk because same return
D. Can't compare without beta

Answer: B. With equal expected returns, larger σ means wider spread and more risk.

3. The coefficient of variation is most useful when…

A. Comparing assets with different expected returns
B. Computing beta
C. Computing the cost of debt
D. All assets have the same return

Answer: A. CV = σ ÷ r̄ is risk per unit of return, fair across different returns.

4. To cut portfolio risk most effectively, combine assets whose correlation is…

A. Perfectly positive (+1)
B. As low as possible, ideally negative
C. Equal to 1.0 like market beta
D. Irrelevant

Answer: B. Low/negative correlation offsets movements; ρ = –1 can even give a risk-free mix.

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5. The risk that CANNOT be diversified away and is therefore relevant is…

A. Unsystematic risk
B. Firm-specific risk
C. Systematic (market) risk
D. Management risk

Answer: C. Systematic risk comes from factors hitting all firms (inflation, rates, recession).

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6. A firm's factory burns down and its stock falls. This is…

A. Systematic
B. Unsystematic / diversifiable
C. Inflation risk
D. Interest-rate risk

Answer: B. A firm-specific event is unsystematic and diversifiable.

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7. A stock has β = 0.5. If the market falls 10%, the stock is expected to…

A. Fall 20%
B. Fall 10%
C. Fall 5%
D. Rise 5%

Answer: C. β 0.5 moves half as much: 0.5 × (–10%) = –5%.

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8. R_F = 2%, r_m = 6%, β = 1.5. The CAPM required return is…

A. 6%
B. 8%
C. 9%
D. 11%

Answer: B. 2% + 1.5 × 4% = 8% (Example 8.14).

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9. On the Security Market Line, the intercept and slope are…

A. β and R_F
B. R_F and the market risk premium ($r_m - R_F$)
C. r_m and β
D. WACC and tax

Answer: B. $r = R_F + \beta(r_m - R_F)$: intercept R_F, slope the market premium.

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10. Expected inflation rises. What happens to the SML?

A. Shifts up in parallel
B. Becomes flatter
C. Shifts down
D. No change

Answer: A. Inflation raises R_F, lifting every required return equally (Fig. 8.10).

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11. Debt costs 6%, equity 14%, 50–50 mix. A 7% project should be…

A. Accepted since 7% > 6%
B. Rejected since 7% < WACC 10%
C. Accepted since debt is cheaper
D. Rejected since 7% < 14%

Answer: B. Use the blended cost: WACC = 10% (Example 9.1).

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12. $1,000 par bond, $60 coupon, 20 years, net proceeds $980. Approximate pre-tax r_d…

A. 6.000%
B. 6.162%
C. 6.500%
D. 4.880%

Answer: B. [60 + 20/20] ÷ 990 = 61 ÷ 990 = 6.162%.

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13. Pre-tax cost of debt 6.177%, tax 21%. After-tax cost?

A. 1.30%
B. 4.88%
C. 6.18%
D. 7.47%

Answer: B. 6.177% × 0.79 = 4.88%; interest is tax-deductible.

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14. Why is debt cost multiplied by (1 – T) but preferred cost is not?

A. Interest is tax-deductible; preferred dividends aren't
B. Preferred is safer than debt
C. Debt has no flotation cost
D. Because of market-value weights

Answer: A. Dividends come from after-tax profit — no tax saving.

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15. 8% preferred, $80 par, $2.50 flotation. Its cost is…

A. 8.000%
B. 8.258%
C. 7.750%
D. 6.400%

Answer: B. 6.40 ÷ 77.50 = 8.258%.

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16. D_1 = $3.99, P_0 = $50, g = 5%. Gordon cost of equity?

A. 7.98%
B. 12.00%
C. 12.98%
D. 13.58%

Answer: C. 0.0798 + 0.05 = 12.98% (Example 9.8).

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17. Why is new common stock (r_n) costlier than retained earnings (r_r)?

A. Underpricing and flotation costs reduce proceeds
B. Retained earnings are free
C. New stock is tax-deductible
D. New stock has a higher beta

Answer: A. $N_n < P_0$ raises the cost; retained earnings still cost r_s.

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18. Weights: 40% debt (4.88%), 10% preferred (8.26%), 50% equity (13%). WACC?

A. 8.71%
B. 9.28%
C. 10.00%
D. 13.00%

Answer: B. 1.95% + 0.83% + 6.50% = 9.28% (Table 9.2).

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19. WACC weights should be based on…

A. Book values
B. Par values
C. Market values (and the target structure)
D. Number of shares

Answer: C. Use market-value weights; the target structure reflects the optimal long-run mix.

Sources

- P3MD Week 8 MK2 — Risk and Return & The Cost of Capital
- Gitman & Zutter — Principles of Managerial Finance, Ch. 8 & 9 (Pearson, 2019)